

A New Floating Active Inductor Using Resistive Feedback Technique

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Abstract — a new floating active inductor is proposed, with resistive feedback technique implemented. Compared to traditional CMOS spiral inductors, the active inductor proposed in this paper can substantially improve its equivalent inductance and quality factor. This resistive feedback floating active inductor was fabricated using the TSMC 0.18 μm RF CMOS process, which demonstrates a maximum quality factor of 68 with a 33 nH inductance. The core area of the proposed active floating inductor is only 0.081 mm² which is one tenth of a traditional spiral inductor. The proposed active inductor also shows more than ten times wide dynamic range and four times higher quality factor compared to conventional floating active inductor circuits.

Index Terms — CMOS, active inductor, floating, resistive feedback.

I. INTRODUCTION

A large range of modern high-volume consumer applications requires the availability of low-power, low-cost, wireless Microsystems. According to these requirements, CMOS technology has become the best choice for wireless communication systems. However, the spiral inductor which is the major passive components in RF integrated circuits can't obtain high quality factor and occupies too large area. Active inductor has attracted much interest in the past few years [1]-[6]. Compared to typical on chip spiral inductor, the advantages of active inductor are large inductance value, high quality factor, small chip area, tunable inductance and quality factor and the possibility of achieving higher inductance with high resonance frequency. Application of the active inductor includes Wilkinson power divider, phase shifter, Q filter, oscillator, and current-mode phase-locked loop.

However, most of the previously proposed active inductors are originally grounded-type 1-port network. When the 2-port floating characteristics are required, the grounded node is usually floated simply by additional current source and bypass capacitor. These floating active inductors don't have symmetric structure, they show somewhat different characteristics from each port, which deviates from the behavior of an ideal inductor.

In this paper we proposed a symmetric floating active inductor. It is composed of a basic differential-pair transconductors and a feedback resistor. The feedback resistor was added between the two transconductors of the active inductor to increase the inductance and at the same time to lower the parasitic series resistance, thereby boosting the quality factor of active inductor. The paper is organized as follow, Section II describe the conventional feedback

resistance floating active theory. The theoretical analysis and circuit design of the proposed new resistive feedback floating active inductor are presented in Section III. The measurements results of the proposed circuits are shown in Section IV. Conclusion is given in Section V.

II. THEORY OF FEEDBACK RESISTANCE FLOATING ACTIVE INDUCTORS

In the literature, both non-feedback [3] and feedback active floating inductor [4-6] have been implemented. The schematic of the active inductor proposed in [3] is shown in Fig. 1. It is a differentially configured gyrator-C floating inductor. Transistors M5 and M6 are biased in the triode region and behaved as voltage-controlled resistors whose value are controlled by the gate voltage V_b . All other transistors are biased in the saturation.

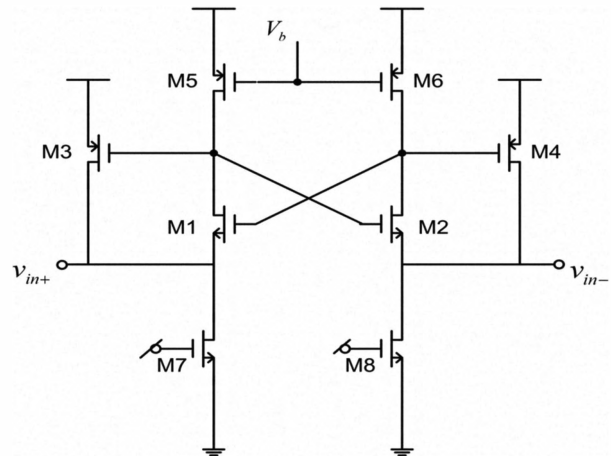


Fig. 1 Schematic of non-feedback active inductor.

Derived from small signal equivalent circuit of the active inductor, the differential input impedance is given in formula (1)

$$Z = \frac{2[s(C_{gs1} + C_{gs3}) - g_{m1} + g_{ds5}]}{g_{ds5}[g_{m1} + g_{m3} + s(C_{gs1} + C_{gs3})]} \quad (1)$$

The parameter of the RLC equivalent circuit of this active inductor are given by

$$R_p = \frac{2}{g_{ds5}},$$

$$R_s = \frac{2(g_{ds5} - g_{m1})}{g_{ds5}(2g_{gm1} + g_{m3} - g_{ds5})}, \quad (2)$$

$$L = \frac{2(C_{gs1} + C_{gs3})}{g_{ds5}(2g_{m1} + g_{m3} - g_{ds5})}$$

Note that $g_{m1} < g_{ds5} < 2g_{m1} + g_{m3}$ are required in order for R_s and L to have a positive value. The drawback of this active is that the value of L and Q is small.

In [4], a resistance feedback active was proposed based on [3]. The resistor is added into the cross-coupling trans conductance to increase the value of inductance and quality factor. However, both of these two floating active inductance occupy too much chip area. In [5], [6] the Q value can achieve up to 100, but this is realized by using double number transistors. Its area is even bigger than the traditional spiral inductor.

III. THEORY ANALYSIS AND CIRCUIT DESIGN OF PROPOSED FLOATING ACTIVE INDUCTOR

Fig. 2 shows the schematic of the proposed floating active inductor. M1 and M4 are PMOS transistors, M3 and M6 are NMOS transistors, they all connected in common-source mode to realize bias current source. M2 and M5 conform a cross coupled pair which can produce inductance impedance. Feedback resistor is added between the cross coupled transistors M2 and M5.

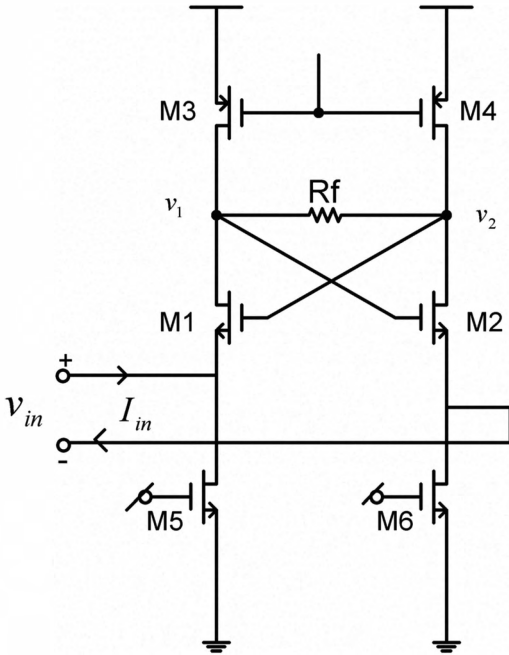


Fig. 2 The schematic of proposed active inductor

Fig.3 shows the small signal model of proposed active inductor.

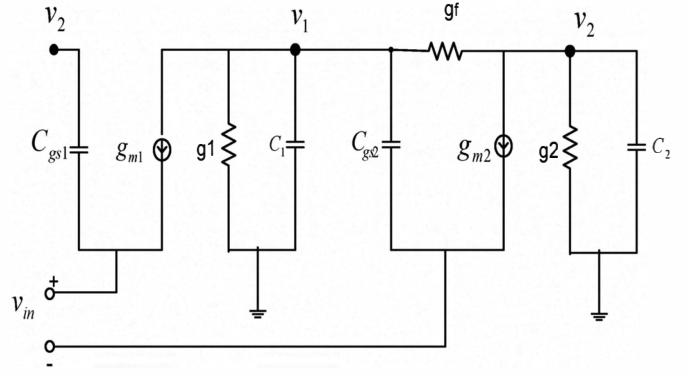


Fig. 3 Small signal mode of proposed active inductor

In figure 3, g_f is the conductance of feedback resistor R . C_1, g_1, C_2, g_2 are the effective parasitic capacitance and conductance at node v_1 and v_2 .

Assuming that the operating frequency is much lower than MOSFET's cutoff frequency ($g_m \gg |sC_{gs}|$), then the input impedance of the floating active inductor in Fig.1 can be derived as follows:

$$Y_{in} = g_{m1} + \frac{g_{m1}^2(g_{m2} - g_f)}{(g_1 + g_f + sC_1)(g_1 + g_f + sC_2) + (g_{m1} - g_f)(g_f - g_{m2})} \quad (3)$$

Fig. 4 shows the RLC equivalent circuit of the proposed active inductor.

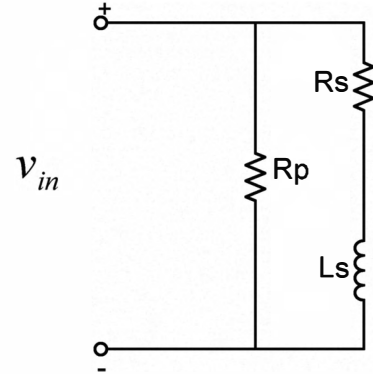


Fig. 4 The equivalent RLC circuit of proposed inductor

To find out the parameters of the RLC equivalent circuit of the active inductor, we transform Equation (3) to (4).

$$Y_{in}(j\omega) = g_{m1} + 1 \left[\frac{(g_1 + g_f)(g_2 + g_f) - \omega^2 C_1 C_2 + (g_{m1} - g_f)(g_f - g_{m2})}{g_{m1}^2 (g_{m2} - g_f)} + \frac{j\omega(C_1(g_2 + g_f) + C_2(g_1 + g_f))}{g_{m1}^2 (g_{m2} - g_f)} \right] \quad (4)$$

As can be seen in Equation (4), the parameters of the RLC equivalent circuit of the active inductor are given by

$$\begin{aligned} R_p &= \frac{1}{g_{m1}} \\ L_s &= \frac{C_1(g_2 + g_f) + C_2(g_1 + g_f)}{g_{m1}^2 (g_{m2} - g_f)} \\ R_s &= \frac{(g_1 + g_f)(g_2 + g_f) - \omega^2 C_1 C_2 + (g_{m1} - g_f)(g_f - g_{m2})}{g_{m1}^2 (g_{m2} - g_f)} \end{aligned} \quad (5)$$

From Equation (5), it is observed that feedback resistor introduce a minus item in the denominator of equivalent inductance expression, if the value of g_{m2} and g_f is chosen to be closely, the inductance of the proposed floating active inductor will be high.

The quality factor Q of an inductor quantifies the ratio of the net magnetic energy stored in the inductor to its ohm loss in one oscillation cycle. The Q of active inductor can be quantified as follow. [11]

$$Q = \frac{\text{Im}[Z]}{\text{Re}[Z]} \quad (6)$$

The quality factor of the proposed active inductor can be derived directly from Equation (4)

$$Q = \left(\frac{\omega L}{R_s} \right) \frac{R_p}{R_p + R_s \left[1 + \left(\frac{\omega L}{R_s} \right)^2 \right]} \quad (7)$$

It is seen that the first term in (7), denoted by

$$Q_1 = \frac{\omega L}{R_s} \quad (8)$$

Quantifies the quality factor of the active inductor, whereas the second term, denoted by

$$Q_2 = \frac{R_p}{R_p + R_s \left[1 + \left(\frac{\omega L}{R_s} \right)^2 \right]} \quad (9)$$

Accounts for the effect of the finite output impedance of the deep sub-micro MOSFETs.

It is seen that Q_1 dominates the quality factor of active inductor. To boost the quality factor of active inductors, R_s must be minimized. However, it is can be seen in Equation (5)

that $g_{m2} - g_f$ item is also appeared in the denominator of equivalent series resistor. In order to realize smaller series resistor R_s while maintain higher value of active inductor, the value of nominator of series resistor is design to be close to zero by optimizing circuit parameter and the denominator of series resistor is design to be as small as possible.

IV. EXPERIMENTAL RESULTS

The proposed floating active inductor was fabricated using the TSMC 0.18um CMOS process. Fig. 5 shows the die photo with chip size of $0.4mm \times 0.4mm$ include all the test pads. On wafer probing was used to characterize two port S-parameters of the proposed floating active inductor. Equivalent inductor is calculated by using following formula [7]:

$$Z_{in} = Z_{11} - Z_{21} - Z_{12} + Z_{22} \quad (10)$$

$$L = \frac{\text{Im}(Z_{in})}{\omega} \quad (11)$$

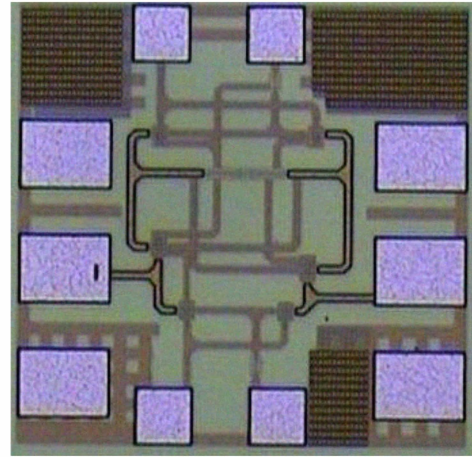


Fig. 5 The die photo of the fabricated active inductor

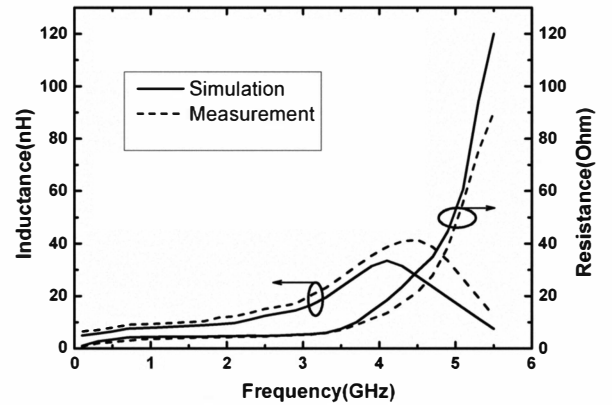


Fig. 6 Simulated and measured resistance and inductance of the proposed active inductor

Fig. 6 shows the simulated and measured equivalent series inductance and resistance. The measured quality factor and inductances are shown in Fig. 7

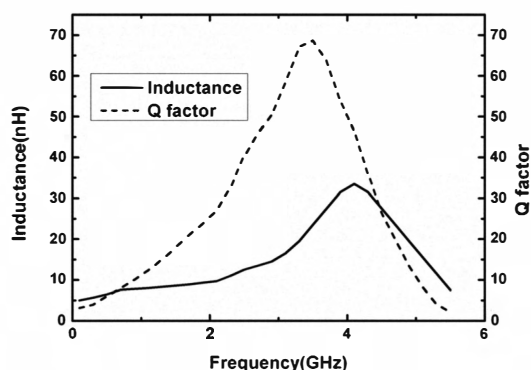


Fig. 7 Measured Q and inductance of proposed active inductor

As can be seen in Fig. 7, the maximum measured quality factor is about 68 at 3.6 GHz and the maximum measured inductance is 33 nH at 4 GHz. As the frequency increase, the inductance of proposed active inductor will become to capacitive because of the capacitor at the input node.

V. CONCLUSION

A new floating active inductor using resistive feedback technique has been proposed in this paper. The new proposed active inductor uses a minimum number of transistors to construct the cross coupled gyrator. The maximum Q factor is 68 and maximum inductance is 33 nH. The total power consumption is 3.6 mW. This configuration can be easily implemented using the series circuit.

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